



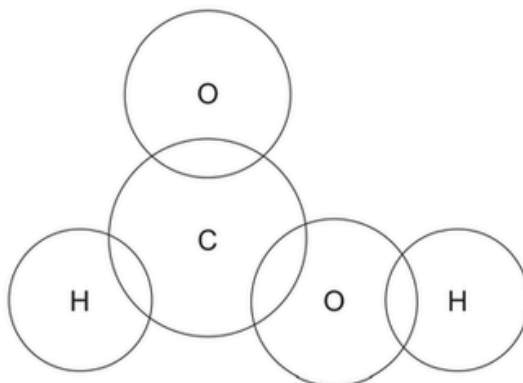
AS level Chemistry

# CHEMICAL BONDING

THEORY

2026 EDITION

- (d) Complete the 'dot-and-cross' diagram, showing outer electrons only, to show the bonding in methanoic acid,  $\text{HCO}_2\text{H}$ .



[2]

[Total: 9]

**5** Hydrocarbon molecules contain covalent bonds.

**(a)** Define covalent bond.

.....  
..... [1]

**(i)** Define covalent bond.

.....  
..... [1]

- (iii)  $\text{POCl}_3$  contains a double covalent bond between P and O.

Complete the dot-and-cross diagram, in Fig. 1.1, to show the bonding in  $\text{POCl}_3$ .

Show outer shell electrons only.

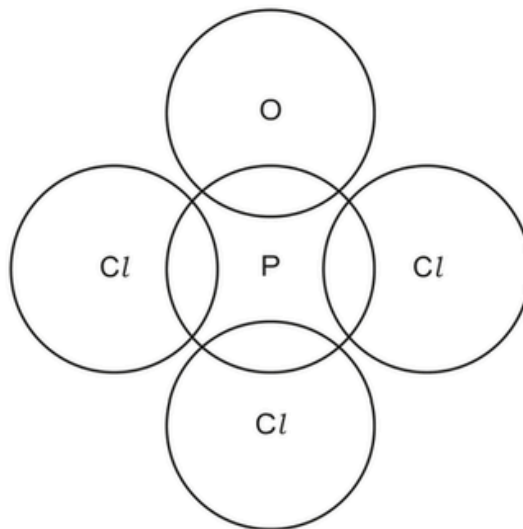


Fig. 1.1

[2]

Q5

9701/22/M/J/21/Q2(b)(i)

Dative Bonding

(b) Carbon monoxide, CO, is a gas at room temperature and pressure. It contains a coordinate bond.

(i) Explain what is meant by *coordinate bond*.

.....  
 ..... [1]

Q6

9701/23/M/J/22/Q3(a)(i)

Dative Bonding

(i) The atoms in a molecule of CO are held together by a triple covalent bond. One of these bonds is a coordinate (dative covalent) bond.

Draw a dot-and-cross diagram to show the arrangement of outer electrons in a CO molecule.

Use • to represent electrons from an oxygen atom.

Use × to represent electrons from a carbon atom.

[1]

Q7

9701/22/F/M/21/Q2(e)(i)

Dative Bonding

(e) Aluminium reacts with chlorine to form aluminium chloride.

Aluminium chloride can exist as the gaseous molecule  $Al_2Cl_6(g)$ . This molecule contains coordinate bonds.

(i) Draw a diagram that clearly shows all the types of bond present in  $Al_2Cl_6(g)$ .

[2]

Q8

9701/22/M/J/21/Q2(b)(ii)

Dative Bonding

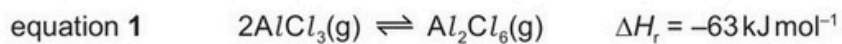
(ii) Draw a 'dot-and-cross' diagram to show the arrangement of outer electrons in CO.

Show the electrons belonging to the C atom as ×.

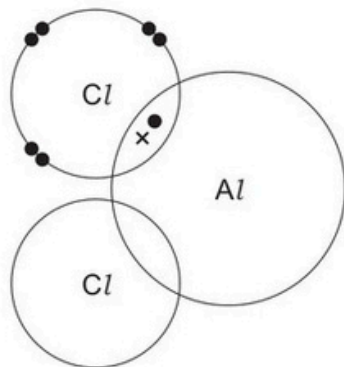
Show the electrons belonging to the O atom as ●.

[2]

(c) In the gas phase,  $AlCl_3(g)$  exists at equilibrium with  $Al_2Cl_6(g)$  as shown.

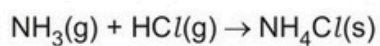


(i) Complete the dot-and-cross diagram to show the bonding in  $Al_2Cl_6$ .



[2]

4 (a)  $NH_3(g)$  reacts with  $HCl(g)$  to produce  $NH_4Cl(s)$ , as shown.



Draw a diagram to show the ionic, covalent and coordinate bonding present in a formula unit of  $NH_4Cl$ .

[2]

Q11

9701/22/O/N/22/Q1(a)

Ionic Bonding

1 Species such as  $\text{NH}_4^+$ ,  $\text{CO}_3^{2-}$  and  $\text{PO}_4^{3-}$  are examples of molecular ions.

(a) Ionic and covalent bonds both involve an electrostatic attraction between different species.

Identify the species that are electrostatically attracted to one another in:

- an ionic bond

.....

- a covalent bond.

.....

[2]

Q12

9701/22/M/J/22/Q2(a)-(b)

Ionic Bonding

2 Radium, Ra, is an element found in Group 2 of the Periodic Table. It is a crystalline solid at room temperature and conducts electricity.

(b) Draw a dot-and-cross diagram to show the arrangement of outer electrons in  $\text{RaCl}_2$ .

[1]

**(b) (i)** LiH is an ionic compound.

Draw a dot-and-cross diagram of LiH.

Include **all** electrons.

[2]

(a) Lithium sulfide,  $\text{Li}_2\text{S}$ , is a crystalline solid with a melting point of  $938^\circ\text{C}$ . It conducts electricity when it is molten.

(i) Give the formulae of the particles present in solid lithium sulfide.

..... [1]

1 The melting points of some solids are shown in Table 1.1.

**Table 1.1**

| solid           | melting point / K |
|-----------------|-------------------|
| magnesium       | 923               |
| phosphorus      | 317               |
| sodium chloride | 1074              |
| sulfur          | 392               |

(a) (i) State the type of bonding present in magnesium and in sodium chloride.

bonding in magnesium .....

bonding in sodium chloride .....

[1]

(ii) Explain the difference in the melting points of magnesium and sodium chloride.

.....

..... [1]

- 1 (a) Solid sodium conducts electricity. Sodium oxide conducts electricity when molten but not when solid.

(i) Name the type of bonding present in sodium and in sodium oxide.

bonding in sodium .....

bonding in sodium oxide .....

[1]

(ii) Explain how solid sodium conducts electricity.

.....

..... [1]

(iii) Explain why sodium oxide conducts electricity when molten.

.....

..... [1]

## METALLIC BONDING - EASY

Q17

9701/22/F/M/24/Q1(a)

Metallic Bonding

**1** Bismuth is an element in Group 15 of the Periodic Table.

**(a)** Bismuth has metallic bonding.

Draw a labelled diagram to show the metallic bonding in bismuth.

[1]

Q18

9701/22/F/M/25/Q2(a)

Metallic Bonding

**2 (a)** The Group 2 elements show metallic bonding.

Define metallic bonding.

.....

.....

[2]

Q19

9701/23/M/J/23/Q1(a)(i)-(ii)

Metallic Bonding

**1** Copper is used in electrical equipment. It has a melting point of 1085 °C.

**(ii)** Draw a labelled diagram to show the bonding present in copper.

[1]

Q20

9701/22/F/M/23/Q1(b)(ii)

VSEPR Theory

(ii) Suggest the shape of a molecule of  $\text{H}_2\text{S}$ .

..... [1]

Q21

9701/22/F/M/25/Q1(b)(ii)

VSEPR Theory

(ii) Predict the shape of a molecule of  $\text{PCl}_5$ .

..... [1]

Q22

9701/21/O/N/21/Q1(a)(iii)

VSEPR Theory

(iii) Suggest the bond angle in a molecule of  $\text{CS}_2$ .

..... [1]

## VSEPR THEORY - EASY (CONT.)

Q23

9701/22/O/N/24/Q3(b)(iv)

VSEPR Theory

(iv) Suggest the shape of the  $[\text{Al}(\text{OH})_4]^-$  ion.

..... [1]

Q24

9701/21/M/J/24/Q2(b)(ii)

VSEPR Theory

(ii) Predict the shape of, and bond angle in, a molecule of  $\text{SCl}_2$  by using VSEPR theory.

shape .....

bond angle .....

[2]

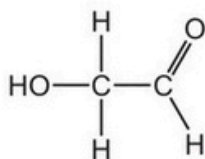
Q25

9701/22/F/M/21/Q4(a)

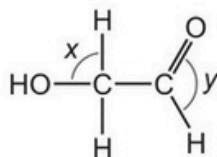
VSEPR Theory

- 4 Hydroxyethanal,  $\text{HOCH}_2\text{CHO}$ , has been observed in dust clouds near the centre of our galaxy.

hydroxyethanal



- (a) Predict the bond angles labelled  $x$  and  $y$  in the diagram of hydroxyethanal.



$x = \dots\dots\dots^\circ$

$y = \dots\dots\dots^\circ$   
[2]

Q26

9701/21/M/J/23/Q1(e)

VSEPR Theory

- (e) Te reacts with  $\text{F}_2$  at  $150^\circ\text{C}$  to form  $\text{TeF}_x$ . Molecules of  $\text{TeF}_x$  are octahedral with bond angles of  $90^\circ$ .

Explain why  $\text{TeF}_x$  is octahedral with bond angles of  $90^\circ$ .

.....  
 .....  
 ..... [2]

Q27

9701/21/O/N/22/Q3(d)(v)

VSEPR Theory

- (v) Suggest a value for the  $\text{Cl-S-S}$  bond angle in isomer I. Explain your answer.

bond angle =  $\dots\dots\dots^\circ$

explanation .....  
 .....  
 .....  
 ..... [2]

(b) Fig. 4.2 shows a possible arrangement of outer-shell electrons in one  $\text{SO}_2$  molecule.

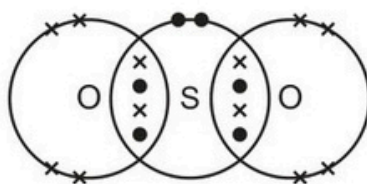


Fig. 4.2

(i) Use Fig. 4.2 to predict the shape and bond angle of a molecule of  $\text{SO}_2$ .

shape .....

bond angle .....<sup>o</sup>

[2]

(v)  $\text{I}_2$  reacts with  $\text{NH}_3$  to form  $\text{NI}_3$ .

Predict the shape of a molecule of  $\text{NI}_3$ . Explain your answer.

shape .....

explanation .....

.....

[2]

- 1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

**Table 1.1**

| element                                         | H    | Li   | C    | O    | S    |
|-------------------------------------------------|------|------|------|------|------|
| Pauling electronegativity value                 | 2.1  | 1.0  | 2.5  | 3.5  | 2.6  |
| first ionisation energy / $\text{kJ mol}^{-1}$  | 1310 | 519  | 1090 | 1310 | 1000 |
| second ionisation energy / $\text{kJ mol}^{-1}$ | —    | 7300 | 2350 | 3390 | 2260 |

- (a) (i) Define electronegativity.

.....  
..... [1]

## ELECTRONEGATIVITY - HARD

Q31

9701/22/F/M/23/Q1(a)(ii)

Electronegativity

- (ii) O and S are in Group 16.

Explain the difference in the Pauling electronegativity values of O and S.

.....

.....

..... [2]

Q32

9701/22/M/J/23/Q1(b)(ii)

Electronegativity

- (ii) Explain why electronegativity increases across a period.

.....

.....

..... [2]

Q33

9701/22/M/J/21/Q2(c)(iv)

Electronegativity

- (iv) Suggest why the bond in a molecule of CO contains a dipole moment.

..... [1]

Q34

9701/22/F/M/23/Q1(e)(i)

Electronegativity

- (e) (i) Complete Table 1.2 by placing a tick (✓) to show which of the compounds have molecules with an overall dipole moment.

Table 1.2

|                       |       |       |       |       |
|-----------------------|-------|-------|-------|-------|
| compound              | O=C=O | O=S=O | S=C=S | S=C=O |
| overall dipole moment |       |       |       |       |

[2]

Q35

9701/23/M/J/25/Q4(b)(ii)

Electronegativity

- (ii) Use Table 4.1 to predict the strength of the dipole moment of SO<sub>2</sub>, if any, compared to that of H<sub>2</sub>O. Explain your answer.

Table 4.1

|                   |     |     |     |
|-------------------|-----|-----|-----|
|                   | H   | O   | S   |
| electronegativity | 2.1 | 3.5 | 2.6 |

.....

.....

..... [2]

- (c) Complete the table to show the numbers of sigma ( $\sigma$ ) bonds and pi ( $\pi$ ) bonds present in propene,  $C_3H_6$ , and  $C_3H_8O$ .

|           | $\sigma$ | $\pi$ |
|-----------|----------|-------|
| $C_3H_6$  |          |       |
| $C_3H_8O$ |          |       |

[2]

- (b) Fig. 5.2 shows two reactions of **W** to produce organic compounds **Y** and **Z**.

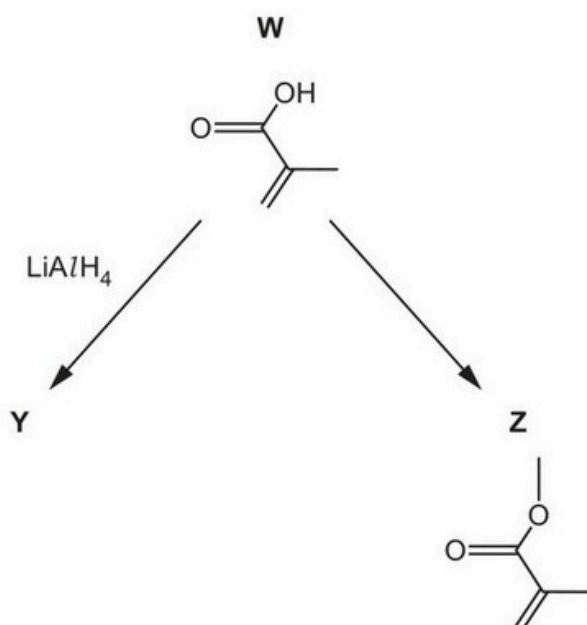


Fig. 5.2

- (i) Deduce the number of sigma ( $\sigma$ ) bonds and pi ( $\pi$ ) bonds present in **Z**.

number of  $\sigma$  bonds .....

number of  $\pi$  bonds .....

[2]

(iv) Complete Fig. 3.2 to show the pi ( $\pi$ ) bond in **L** that is formed from orbital overlap.

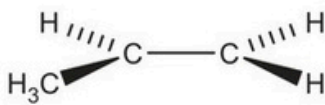


Fig. 3.2

[1]

(ii) Ethyne, HCN and **Q** all contain triple bonds between two atoms.

A triple bond consists of one sigma ( $\sigma$ ) and two pi ( $\pi$ ) bonds.

Draw a labelled diagram to show the formation of one pi ( $\pi$ ) bond.

[2]

- (iii) Draw a diagram of the  $\pi$  bond between the N atoms in  $\text{N}_2\text{F}_2$  and describe how it forms.

.....  
 ..... [2]

- (d) Describe the covalent bonds present between the carbon atoms in an ethene molecule by completing Table 4.2.

Table 4.2

|                                   | sigma ( $\sigma$ ) | pi ( $\pi$ ) |
|-----------------------------------|--------------------|--------------|
| type of orbitals involved in bond |                    |              |
| how the orbitals overlap          |                    |              |

[2]

- (ii) Draw labelled diagrams to show, in terms of orbital overlap, how the  $\sigma$  and  $\pi$  bonds are made in a C=C bond.

$\sigma$  bond

$\pi$  bond

[2]

- (c) In electrophilic reactions involving alkenes the  $\pi$  bond of C=C is broken.

- (i) Suggest **one** difference between  $\sigma$  and  $\pi$  bonds that explains why the  $\pi$  bond is broken in electrophilic addition reactions involving alkenes.

.....

..... [1]

- (f) At very high temperatures, phosphorus can form  $P_2$  molecules.

$P_2$  contains a triple bond,  $P \equiv P$ .

- (i) Describe the formation of the  $P \equiv P$  bond in terms of orbital overlap.

.....

.....

.....

..... [2]

Q44

9701/22/F/M/24/Q3(d)(ii)

Hybridisation

(ii) Deduce the hybridisation of the N atoms in  $\text{N}_2\text{F}_2$ .

..... [1]

Q45

9701/21/M/J/24/Q5(b)(i)

Hybridisation

(b) A  $\text{C}=\text{C}$  bond in an alkene is made from a  $\sigma$  bond and a  $\pi$  bond.

(i) Identify the hybridisation of the carbon atoms in a  $\text{C}=\text{C}$  bond in an alkene.

..... [1]

Q46

9701/22/M/J/25/Q4(c)(i)

Hybridisation

(c) Compounds **A**, **B** and **C** belong to the alcohol homologous series. Each molecule of **A**, **B** and **C** contains four saturated carbon atoms.

(i) Identify the type of hybridisation shown in the saturated carbon atoms of all alcohols.

..... [1]

(c) In reaction 4, calcium carbide,  $\text{CaC}_2$ , is formed from  $\text{CaO}$ .

$\text{CaC}_2$  contains the  $\text{C}_2^{2-}$  anion. Each carbon in  $\text{C}_2^{2-}$  is sp hybridised.

(i) Describe how sp hybridised orbitals are formed.

.....  
..... [1]

(ii) Sketch a diagram to show how two sp hybrid orbitals can form a sigma ( $\sigma$ ) bond.

[2]

Q48

9701/21/M/J/22/Q2(a)

Hybridisation

2 Nitrogen molecules,  $\text{N}_2(\text{g})$ , contain two atoms attracted to each other by a triple covalent bond.

- (a) Describe how the triple covalent bond forms in a  $\text{N}_2(\text{g})$  molecule. Refer to orbital overlap and hybridisation in your answer.

.....

.....

.....

..... [3]

Q49

9701/21/M/J/23/Q4(b)(iii)

Hybridisation

- (iii) Complete Table 4.1 to show the number of  $\text{sp}^2$  and  $\text{sp}^3$  hybridised carbon atoms that are present in a molecule of **V**.

Table 4.1

| type of hybridisation              | $\text{sp}^2$ | $\text{sp}^3$ |
|------------------------------------|---------------|---------------|
| number of carbon atoms in <b>V</b> |               |               |

[2]

Q50

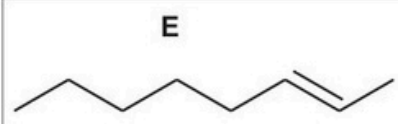
9701/22/M/J/24/Q4(e)(i)

Hybridisation

- (e) (i) Both  $\sigma$  and  $\pi$  bonds are present in a molecule of **E** as a result of different types of hybridisation in the carbon atoms.

Complete Table 4.2 to show the number of carbon atoms with each type of hybridisation in a molecule of **E**.

Table 4.2

|                                                                                                     | number of carbon atoms |                          |                          |
|-----------------------------------------------------------------------------------------------------|------------------------|--------------------------|--------------------------|
|                                                                                                     | $\text{sp}$ hybridised | $\text{sp}^2$ hybridised | $\text{sp}^3$ hybridised |
| <p><b>E</b></p>  |                        |                          |                          |

[2]

Q51

9701/22/M/J/21/Q2(c)(iii)

Intermolecular Forces

- (iii) Complete the table by naming **all** the types of intermolecular forces (van der Waals') in separate samples of  $\text{N}_2(\text{g})$  and  $\text{CO}(\text{g})$ .

|                                        | $\text{N}_2(\text{g})$ | $\text{CO}(\text{g})$ |
|----------------------------------------|------------------------|-----------------------|
| number of electrons per molecule       | 14                     | 14                    |
| presence of a dipole moment            | $\times$               | $\checkmark$          |
| boiling point/ $^{\circ}\text{C}$      | -195.8                 | -191.5                |
| intermolecular forces (van der Waals') |                        |                       |

[2]

Q52

9701/22/F/M/25/Q3(d)(i)

Intermolecular Forces

- (d) Iodine monobromide,  $\text{IBr}$ , is a dark red solid that melts near room temperature.

$\text{IBr}$  reacts with propene. The mechanism for this reaction is the same as the mechanism for that of  $\text{HBr}$  with propene.

- (i) Identify **all** the intermolecular forces that exist between molecules of  $\text{IBr}$ .

.....  
 ..... [1]

(iii) Name the strongest intermolecular force that exists between  $\text{NH}_3(\text{l})$  molecules.

..... [1]

Q54

9701/23/M/J/21/Q2(a)

Intermolecular Forces

**2** Methanol,  $\text{CH}_3\text{OH}$ , is soluble in water because it forms hydrogen bonds with water molecules.

- (a)** Draw a fully labelled diagram to show how a hydrogen bond forms between a water molecule and a methanol molecule.

[3]

Q55

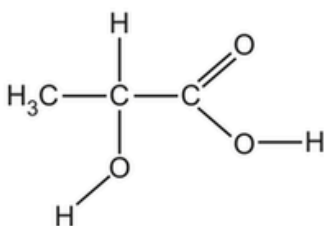
9701/22/O/N/23/Q4(a)(iii)

Intermolecular Forces

**(iii)** Lactic acid forms hydrogen bonds with water.

Complete Fig. 4.2 to show the formation of a hydrogen bond between one molecule of lactic acid and one molecule of water.

Label the hydrogen bond. Show any relevant dipoles and lone pairs of electrons.



**Fig. 4.2**

[3]

- (iv) Draw a diagram to show the formation of the strongest intermolecular force between **two** molecules of  $\text{NH}_3(\text{l})$ .

Include any relevant lone pairs of electrons and dipoles.

[2]

- (v) The melting points of ice and ammonia are shown in Table 1.2.

Table 1.2

| solid   | melting point/K |
|---------|-----------------|
| ice     | 273             |
| ammonia | 195             |

Suggest **two** reasons for the difference in the melting points of ice and ammonia.

.....  
.....  
..... [2]

## INTERMOLECULAR FORCES - HARD (CONT.)

Q57

9701/22/O/N/22/Q2(e)(ii)

Intermolecular Forces

- (ii)  $\text{SCl}_2$  has  $M_r = 103.1$  and is a liquid at room temperature.  $\text{SBr}_2$  has  $M_r = 191.9$  and is a gas at room temperature.

Explain the difference in the physical state of  $\text{SCl}_2$  and  $\text{SBr}_2$ . Give your answer in terms of intermolecular forces.

.....

.....

.....

..... [2]

Q58

9701/23/M/J/21/Q2(b)(iii)

Intermolecular Forces

- (iii) Suggest and explain why the vapour pressure of water at room temperature is lower than the vapour pressure of methanol at room temperature. Refer to the correct intermolecular forces in your answer.

.....

.....

..... [2]

## INTERMOLECULAR FORCES - HARD (CONT.)

Q59

9701/21/O/N/25/Q1(c)(iv)

Intermolecular Forces

- (iv) Methanal is a liquid at  $-30^{\circ}\text{C}$  but carbon dioxide is a gas at this temperature. Explain why.

.....

.....

.....

..... [2]

(d) Fig. 3.2 shows the boiling points of the simplest hydrides of the Group 14 elements, C to Pb.

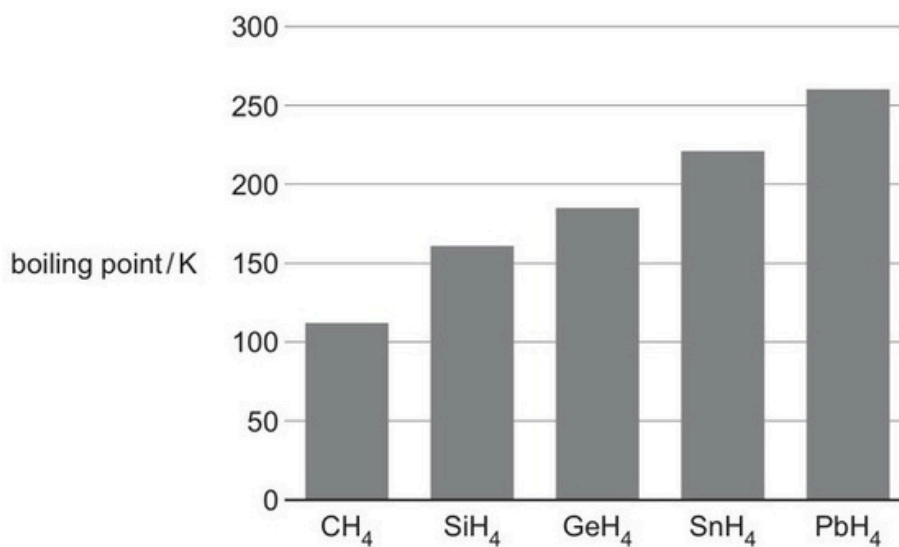


Fig. 3.2

(i) Explain the trend in the boiling points of the Group 14 hydrides shown in Fig. 3.2.

.....

.....

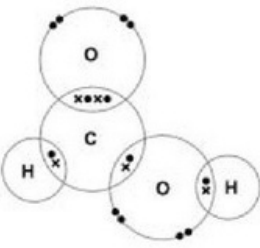
.....

..... [2]

Q1

9701/21/M/J/21/Q4(d)

Covalent Bonding

|      |                                                                                                                       |   |
|------|-----------------------------------------------------------------------------------------------------------------------|---|
| 4(d) |  <p>M1 correct bonding electrons</p> | 1 |
|      | M2 correct number of non-bonding electrons around each oxygen                                                         | 1 |

Q2

9701/21/M/J/24/Q5(a)

Covalent Bonding

|      |                                                                                     |   |
|------|-------------------------------------------------------------------------------------|---|
| 5(a) | (electrostatic) attraction between nuclei of two atoms and shared pair of electrons | 1 |
|------|-------------------------------------------------------------------------------------|---|

Q3

9701/24/M/J/25/Q3(d)(i)

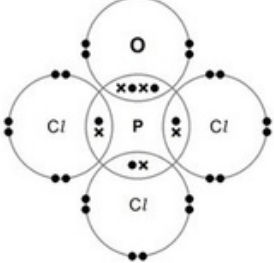
Covalent Bonding

|         |                                                                                             |   |
|---------|---------------------------------------------------------------------------------------------|---|
| 3(d)(i) | (electrostatic) attraction between the nuclei (of two atoms) and a shared pair of electrons | 1 |
|---------|---------------------------------------------------------------------------------------------|---|

Q4

9701/21/O/N/23/Q1(d)(iii)

Covalent Bonding

|           |                                                                                   |   |
|-----------|-----------------------------------------------------------------------------------|---|
| 1(d)(iii) |  | 1 |
|           | Correct bonding e <sup>-</sup>                                                    |   |
|           | rest of electrons <b>correct</b> (i.e. to make 32 electrons in total)             | 1 |

Q5

9701/22/M/J/21/Q2(b)(i)

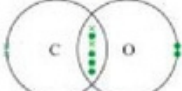
Dative Bonding

|         |                                                                                                                                                                             |   |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 2(b)(i) | (covalent) bond with both electrons are provided from the same / one species<br>OR<br>shared pair (of electrons) are provided from the same species / one atom <i>owtte</i> | 1 |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

Q6

9701/23/M/J/22/Q3(a)(i)

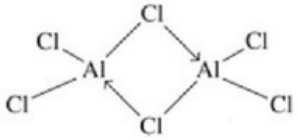
Dative Bonding

|         |                                                                                     |   |
|---------|-------------------------------------------------------------------------------------|---|
| 3(a)(i) |  | 1 |
|---------|-------------------------------------------------------------------------------------|---|

Q7

9701/22/F/M/21/Q2(e)(i)

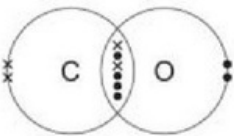
Dative Bonding

|         |                                                                                                                                                                            |   |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 2(e)(i) |  <p><b>M1:</b> 2 × coordinate bonds in the right place<br/><b>M2:</b> all other bonds</p> | 2 |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

Q8

9701/22/M/J/21/Q2(b)(ii)

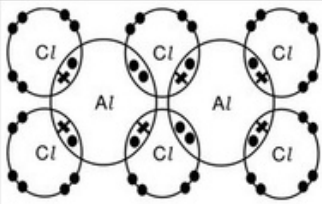
Dative Bonding

|          |                                                                                                                                                                                               |   |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 2(b)(ii) | <p>3 bonding pairs between C and O, 4 •'s AND 2 ×'s<br/>1 lone pair on C, ××, AND 1 lone pair on O, ••.</p>  | 2 |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

Q9

9701/22/O/N/22/Q2(c)(i)

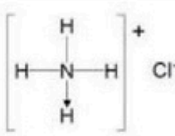
Dative Bonding

|         |                                                                                                                                                                                                                                                                                                      |   |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 2(c)(i) |                                                                                                                                                                                                                    | 1 |
|         | <p>'dot-and-cross' diagram to show<br/><b>M1</b> correct number and connectivity of Cl and Al atoms<br/><b>M2</b> 2 correct dative bonds (→) between Cl and Al of AlCl<sub>3</sub> neighbour <b>AND</b> correct number of non-bonding electrons around each atom of Al<sub>2</sub>Cl<sub>6</sub></p> | 1 |

Q10

9701/23/M/J/24/Q4(a)

Dative Bonding

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 4(a) |  <p>OR</p> <p>Correct representation or label showing<br/><b>M1</b> (ionic bond between) NH<sub>4</sub><sup>+</sup> and Cl<sup>-</sup> <b>AND</b> 4 N—H covalent bonds OR 4 shared pairs of electrons in NH<sub>4</sub><br/><b>M2</b> 1 N—H coordinate bond OR arrow on bond/line in correct direction or only 1 bonding pair using the same symbol for electrons</p> | 2 |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

|     |                      |               |
|-----|----------------------|---------------|
| Q11 | 9701/22/O/N/22/Q1(a) | Ionic Bonding |
|-----|----------------------|---------------|

|      |                                                                 |   |
|------|-----------------------------------------------------------------|---|
| 1(a) | M1 ionic = oppositely charged ions / cations <b>AND</b> anions  | 1 |
|      | M2 covalent = shared pair(s) of electrons <b>AND</b> two nuclei | 1 |

|     |                          |               |
|-----|--------------------------|---------------|
| Q12 | 9701/22/M/J/22/Q2(a)-(b) | Ionic Bonding |
|-----|--------------------------|---------------|

|      |                                                                                                                 |   |
|------|-----------------------------------------------------------------------------------------------------------------|---|
| 2(a) |                                                                                                                 |   |
| 2(b) | <p>Ra<sup>2+</sup> and 2 x Cl<sup>-</sup> 0 electrons surrounding 2Ra <b>AND</b> 8 electrons surrounding Cl</p> | 1 |

|     |                         |               |
|-----|-------------------------|---------------|
| Q13 | 9701/22/F/M/23/Q1(b)(i) | Ionic Bonding |
|-----|-------------------------|---------------|

|         |  |   |
|---------|--|---|
| 1(b)(i) |  | 2 |
|---------|--|---|

|     |                              |               |
|-----|------------------------------|---------------|
| Q14 | 9701/22/M/J/21/Q2(a)(i)-(ii) | Ionic Bonding |
|-----|------------------------------|---------------|

|         |                                     |   |
|---------|-------------------------------------|---|
| 2(a)(i) | Li <sup>+</sup> AND S <sup>2-</sup> | 1 |
| 2(a)(i) |                                     |   |

|     |                              |               |
|-----|------------------------------|---------------|
| Q15 | 9701/22/M/J/23/Q1(a)(i)-(ii) | Ionic Bonding |
|-----|------------------------------|---------------|

|          |                                                                               |   |
|----------|-------------------------------------------------------------------------------|---|
| 1(a)(i)  | bonding in magnesium – metallic <b>AND</b> bonding in sodium chloride – ionic | 1 |
| 1(a)(ii) | bonds in NaCl are stronger than bonds in Mg                                   | 1 |

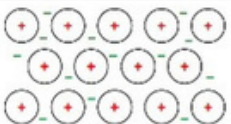
|     |                               |               |
|-----|-------------------------------|---------------|
| Q16 | 9701/21/M/J/25/Q1(a)(i)-(iii) | Ionic Bonding |
|-----|-------------------------------|---------------|

|           |                                                    |   |
|-----------|----------------------------------------------------|---|
| 1(a)(i)   | sodium – metallic <b>AND</b> sodium oxide – ionic  | 1 |
| 1(a)(ii)  | delocalised electrons move (through the structure) | 1 |
| 1(a)(iii) | (sodium and oxide) ions move (through the liquid)  | 1 |

Q17

9701/22/F/M/24/Q1(a)

Metallic Bonding

|      |                                                                                                                                                                                                                                                                            |   |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 1(a) |  <p>labelled diagram shows:</p> <ul style="list-style-type: none"> <li>regular arrangement of positive ions</li> <li><b>AND</b> surrounded by / (sea of) delocalised electrons</li> </ul> | 1 |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

Q18

9701/22/F/M/25/Q2(a)

Metallic Bonding

|      |                                                                                             |   |
|------|---------------------------------------------------------------------------------------------|---|
| 2(a) | electrostatic attraction<br>between positive metal ions / cations and delocalised electrons | 2 |
|------|---------------------------------------------------------------------------------------------|---|

Q19

9701/23/M/J/23/Q1(a)(i)-(ii)

Metallic Bonding

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 1(       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| 1(a)(ii) |  <p>delocalised electrons</p> <p><b>M1</b> diagram showing minimum of 4 particles</p> <ul style="list-style-type: none"> <li>circles containing just a + do not have to be labelled</li> <li>empty circles / circles with Cu must be labelled + ion / positive ion / cation / <math>\text{Cu}^{n+}</math></li> </ul> <p><b>AND</b> with the circles surrounded by electrons shown as <math>e^-</math> / – <b>OR</b> little circles labelled delocalised electrons</p> | 1 |

|          |                          |              |
|----------|--------------------------|--------------|
| Q20      | 9701/22/F/M/23/Q1(b)(ii) | VSEPR Theory |
| 1(b)(ii) | non-linear               | 1            |

|          |                          |              |
|----------|--------------------------|--------------|
| Q21      | 9701/22/F/M/25/Q1(b)(ii) | VSEPR Theory |
| 1(b)(ii) | trigonal bipyramid(al)   | 1            |

|           |                           |              |
|-----------|---------------------------|--------------|
| Q22       | 9701/21/O/N/21/Q1(a)(iii) | VSEPR Theory |
| 1(a)(iii) | 180°                      | 1            |

|          |                          |              |
|----------|--------------------------|--------------|
| Q23      | 9701/22/O/N/24/Q3(b)(iv) | VSEPR Theory |
| 3(b)(iv) | tetrahedral              | 1            |

|          |                                                               |              |
|----------|---------------------------------------------------------------|--------------|
| Q24      | 9701/21/M/J/24/Q2(b)(ii)                                      | VSEPR Theory |
| 2(b)(ii) | <b>M1</b> shape non-linear<br><b>M2</b> bond angle 103–105(°) | 2            |

|      |                                                                                  |              |
|------|----------------------------------------------------------------------------------|--------------|
| Q25  | 9701/22/F/M/21/Q4(a)                                                             | VSEPR Theory |
| 4(a) | <b>M1:</b> $x = 108\text{--}110^\circ$<br><b>M2:</b> $y = 118\text{--}122^\circ$ | 2            |

|      |                                                                                                                                                          |              |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Q26  | 9701/21/M/J/23/Q1(e)                                                                                                                                     | VSEPR Theory |
| 1(e) | <b>M1</b> 6 bonding pairs (of electrons and 0 lone pairs) <b>OR</b> coordination number = 6 <b>OR</b> $x = 6$<br><b>M2</b> bonding pairs repel (equally) | 2            |

|         |                                                                                                                                    |              |
|---------|------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Q27     | 9701/21/O/N/22/Q3(d)(v)                                                                                                            | VSEPR Theory |
| 3(d)(v) | <b>M1</b> $90^\circ < \text{Cl—S—S} \leq 108^\circ$                                                                                | 1            |
|         | <b>M2</b> sulfur has two lone pairs (of e <sup>-</sup> ) (and two bonding pairs)<br><b>AND</b> repulsion from lone pairs (greater) | 1            |

|         |                                                                         |              |
|---------|-------------------------------------------------------------------------|--------------|
| Q28     | 9701/23/M/J/25/Q4(b)(i)                                                 | VSEPR Theory |
| 4(b)(i) | <b>M1</b> non-linear<br><b>M2</b> bond angle – $120 > x \geq 116^\circ$ | 2            |

Q29

9701/24/O/N/25/Q1(b)(v)

VSEPR Theory

|         |                                                                                                         |          |
|---------|---------------------------------------------------------------------------------------------------------|----------|
| 1(b)(v) | <b>M1</b> (trigonal) pyramidal<br><b>M2</b> three bonding pairs <b>AND</b> one lone pair (of electrons) | <b>2</b> |
|---------|---------------------------------------------------------------------------------------------------------|----------|

Q30

9701/22/F/M/23/Q1(a)(i)

Electronegativity

|         |                                                 |          |
|---------|-------------------------------------------------|----------|
| 1(a)(i) | power of an atom to attract electrons to itself | <b>1</b> |
|---------|-------------------------------------------------|----------|

|     |                          |                   |
|-----|--------------------------|-------------------|
| Q31 | 9701/22/F/M/23/Q1(a)(ii) | Electronegativity |
|-----|--------------------------|-------------------|

|          |                                                                                                                                                                                                              |   |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 1(a)(ii) | <ul style="list-style-type: none"> <li>• O lower nuclear charge / lower proton number</li> <li>• O has (one) fewer shell than S / less shielding</li> <li>• greater attraction (for nucleus) in O</li> </ul> | 2 |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

|     |                          |                   |
|-----|--------------------------|-------------------|
| Q32 | 9701/22/M/J/23/Q1(b)(ii) | Electronegativity |
|-----|--------------------------|-------------------|

|          |                                                                                                                                                                                                                                                                                                                                                                      |   |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 1(b)(ii) | (across a period) <ul style="list-style-type: none"> <li>• increase in nuclear charge</li> <li>• similar shielding</li> <li>• (so) increase in nuclear attraction for bonding / outer / valence electrons</li> </ul> <b>OR</b> bonding / outer / valence electron(s) are more strongly attracted to nucleus<br>Two correct for one mark, three correct for two marks | 2 |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

|     |                          |                   |
|-----|--------------------------|-------------------|
| Q33 | 9701/22/M/J/21/Q2(c)(iv) | Electronegativity |
|-----|--------------------------|-------------------|

|          |                                  |   |
|----------|----------------------------------|---|
| 2(c)(iv) | O is more electronegative than C | 1 |
|----------|----------------------------------|---|

|     |                         |                   |
|-----|-------------------------|-------------------|
| Q34 | 9701/22/F/M/23/Q1(e)(i) | Electronegativity |
|-----|-------------------------|-------------------|

|         |                       |       |       |       |       |   |
|---------|-----------------------|-------|-------|-------|-------|---|
| 1(e)(i) | compound              | O=C=O | O=S=O | S=C=S | S=C=O | 2 |
|         | overall dipole moment |       | ✓     |       | ✓     |   |

|     |                          |                   |
|-----|--------------------------|-------------------|
| Q35 | 9701/23/M/J/25/Q4(b)(ii) | Electronegativity |
|-----|--------------------------|-------------------|

|          |                                                                                                                                                                                                                                                              |   |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 4(b)(ii) | <b>M1</b> weaker dipole moment (in SO <sub>2</sub> )<br><b>M2</b> electronegativity (difference) is larger( / 1.4) for H and O than for S and O ( / 0.9)<br><b>OR</b> S=O bond is less polar than H—O<br><b>OR</b> stronger dipoles in S=O / SO <sub>2</sub> | 2 |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

| Question | Answer | Marks |
|----------|--------|-------|
|----------|--------|-------|

|     |                      |            |
|-----|----------------------|------------|
| Q36 | 9701/21/M/J/21/Q6(c) | Sigma Bond |
|-----|----------------------|------------|

|                                 |                                                                                                                                                                                                                                            |       |          |       |                               |   |   |                                 |    |   |   |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------|-------|-------------------------------|---|---|---------------------------------|----|---|---|
| 6(c)                            | <table border="1"> <tr> <td></td><td><math>\sigma</math></td><td><math>\pi</math></td></tr> <tr> <td>C<sub>3</sub>H<sub>6</sub></td><td>8</td><td>1</td></tr> <tr> <td>C<sub>3</sub>H<sub>8</sub>O</td><td>11</td><td>0</td></tr> </table> |       | $\sigma$ | $\pi$ | C <sub>3</sub> H <sub>6</sub> | 8 | 1 | C <sub>3</sub> H <sub>8</sub> O | 11 | 0 | 2 |
|                                 | $\sigma$                                                                                                                                                                                                                                   | $\pi$ |          |       |                               |   |   |                                 |    |   |   |
| C <sub>3</sub> H <sub>6</sub>   | 8                                                                                                                                                                                                                                          | 1     |          |       |                               |   |   |                                 |    |   |   |
| C <sub>3</sub> H <sub>8</sub> O | 11                                                                                                                                                                                                                                         | 0     |          |       |                               |   |   |                                 |    |   |   |

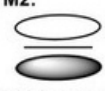
|     |                         |            |
|-----|-------------------------|------------|
| Q37 | 9701/23/M/J/25/Q5(b)(i) | Sigma Bond |
|-----|-------------------------|------------|

|         |                                                            |   |
|---------|------------------------------------------------------------|---|
| 5(b)(i) | number of $\sigma$ bonds = 14<br>number of $\pi$ bonds = 2 | 2 |
|---------|------------------------------------------------------------|---|

|     |                          |            |
|-----|--------------------------|------------|
| Q38 | 9701/24/O/N/25/Q3(a)(iv) | Sigma Bond |
|-----|--------------------------|------------|

|          |                                                                                   |   |
|----------|-----------------------------------------------------------------------------------|---|
| 3(a)(iv) |  | 1 |
|----------|-----------------------------------------------------------------------------------|---|

|     |                          |            |
|-----|--------------------------|------------|
| Q39 | 9701/22/F/M/21/Q3(b)(ii) | Sigma Bond |
|-----|--------------------------|------------|

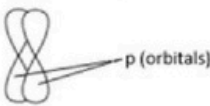
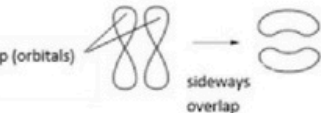
|          |                                                                                                                                                                                                                          |   |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 3(b)(ii) | <p><b>M1:</b> overlap of two p orbitals side-on / above and below the plane</p> <p><b>M2:</b></p>  <p>pi (<math>\pi</math>) orbital</p> | 2 |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

|     |                           |            |
|-----|---------------------------|------------|
| Q40 | 9701/22/F/M/24/Q3(d)(iii) | Sigma Bond |
|-----|---------------------------|------------|

|           |                                                                                                                                       |   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------|---|
| 3(d)(iii) |  <p>M1 shows the pi (<math>\pi</math>) orbital</p> | 1 |
|           | M2 refers to overlap of (two) (2)p orbitals side-on / above and below the plane                                                       | 1 |

|     |                      |            |
|-----|----------------------|------------|
| Q41 | 9701/22/M/J/23/Q4(d) | Sigma Bond |
|-----|----------------------|------------|

|                                                       |                                   |                                     |           |    |   |
|-------------------------------------------------------|-----------------------------------|-------------------------------------|-----------|----|---|
| 4(d)                                                  |                                   |                                     |           |    | 2 |
|                                                       |                                   |                                     | sigma     | pi |   |
|                                                       | type of orbitals involved in bond | sp <sup>2</sup> (-sp <sup>2</sup> ) | (2)p(-2)p |    |   |
|                                                       | how the orbitals overlap          | direct                              | sideways  |    |   |
| Two correct for one mark, four correct for two marks. |                                   |                                     |           |    |   |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 5(b)(ii) | <p><b>M1</b> <math>\sigma</math> bond – diagram showing how <math>\sigma</math> bond is made from direct overlap of orbitals i.e.</p>  <p>OR</p>  <p><b>M2</b> <math>\pi</math> bond - diagram showing how <math>\pi</math> bond is made from sideways overlap of 2 <math>\times</math> p-orbitals i.e.</p>  <p>OR</p>  <p>OR</p>  <p>sideways overlap</p> | 2 |
| 5(c)(i)  | <p><b>EITHER</b> (pair of) electrons in <math>\pi</math> bond are further away from the nuclei so weaker attraction<br/> <b>OR</b> (pair of) electrons in <math>\sigma</math> bond are closer to the two nuclei so stronger attraction</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1 |

|         |                                                                                           |   |
|---------|-------------------------------------------------------------------------------------------|---|
| 3(f)(i) | <b>M1</b> one sigma / $\sigma$ bond <b>AND</b> head-on (overlap of) p / sp (orbitals)     | 1 |
|         | <b>M2</b> two pi / $\pi$ bond(s) <b>AND</b> side-on (overlap of / involving) p (orbitals) | 1 |

|          |        |   |
|----------|--------|---|
| 3(d)(ii) | $sp^2$ | 1 |
|----------|--------|---|

|         |        |   |
|---------|--------|---|
| 5(b)(i) | $sp^2$ | 1 |
|---------|--------|---|

|         |        |   |
|---------|--------|---|
| 4(c)(i) | $sp^3$ | 1 |
|---------|--------|---|

Q47

9701/21/O/N/21/Q2(c)(i)-(ii)

Hybridisation

|          |                                                                                                                                                                                                                                                                                        |   |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 2(c)(i)  | Mixing / overlap / combination of one / an s and one / a p orbital                                                                                                                                                                                                                     | 1 |
| 2(c)(ii) | <p>Sketch a diagram to show HOW two sp hybrid orbitals can form a SIGMA bond</p> <p><b>M1</b> </p> <p><b>M2</b> </p> | 2 |

Q48

9701/21/M/J/22/Q2(a)

Hybridisation

|      |                                                                                                                                                                                                                                                                                                                                             |   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 2(a) | <p><b>M1</b> one sigma / <math>\sigma</math> bond and two pi / <math>\pi</math> bonds</p> <p><b>M2</b> sp hybridisation (in each N atom)</p> <p><b>M3</b> sigma / <math>\sigma</math> forms from direct / head-on / end-on overlap of orbitals</p> <p><b>AND</b> pi / <math>\pi</math> forms sideways / lateral overlap of (p) orbitals</p> | 3 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

Q49

9701/21/M/J/23/Q4(b)(iii)

Hybridisation

|                                    |                                                                                                                                                                                                    |                       |        |        |                                    |   |   |   |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------|--------|------------------------------------|---|---|---|
| 4(b)(iii)                          | <table border="1"> <tr> <td>type of hybridisation</td><td><math>sp^2</math></td><td><math>sp^3</math></td></tr> <tr> <td>number of carbon atoms in <b>V</b></td><td>3</td><td>1</td></tr> </table> | type of hybridisation | $sp^2$ | $sp^3$ | number of carbon atoms in <b>V</b> | 3 | 1 | 2 |
| type of hybridisation              | $sp^2$                                                                                                                                                                                             | $sp^3$                |        |        |                                    |   |   |   |
| number of carbon atoms in <b>V</b> | 3                                                                                                                                                                                                  | 1                     |        |        |                                    |   |   |   |

Q50

9701/22/M/J/24/Q4(e)(i)

Hybridisation

|                                                                                                        |                                                                                                                                                    |                                                                                     |   |   |   |   |
|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---|---|---|---|
| 4(e)(i)                                                                                                | <table><tr><td></td><td>0</td><td>2</td><td>6</td></tr></table> |  | 0 | 2 | 6 | 2 |
|                     | 0                                                                                                                                                  | 2                                                                                   | 6 |   |   |   |
| <p>3 correct answers = 2 marks<br/>2 correct answers = 1 mark<br/>0 or 1 correct answers = 0 marks</p> |                                                                                                                                                    |                                                                                     |   |   |   |   |

Q51

9701/22/M/J/21/Q2(c)(iii)

Intermolecular Forces

|           |                                       |                                                                                  |   |
|-----------|---------------------------------------|----------------------------------------------------------------------------------|---|
| 2(c)(iii) | N <sub>2</sub> (g)                    | CO(g)                                                                            | 2 |
|           | instantaneous dipole–induced dipole ✓ | instantaneous dipole–induced dipole (and)<br>permanent dipole–permanent dipole ✓ |   |

Q52

9701/22/F/M/25/Q3(d)(i)

Intermolecular Forces

|         |                                                                                                                                             |          |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(d)(i) | instantaneous dipole—induced dipole / id—id / London / dispersion (forces)<br><b>AND</b> permanent dipole—permanent dipole / pd—pd (forces) | <b>1</b> |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------|----------|

Q53

9701/22/M/J/23/Q1(b)(iii)

Intermolecular Forces

|           |               |          |
|-----------|---------------|----------|
| 1(b)(iii) | hydrogen bond | <b>1</b> |
|-----------|---------------|----------|

Q54

9701/23/M/J/21/Q2(a)

Intermolecular Forces

|      |                                                                                                                                                                                                                            |          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(a) | <b>M1</b> H-bond between an O and an H in OH groups in water and methanol molecules                                                                                                                                        | <b>1</b> |
|      | <b>M2</b> minimum three partial charges (in sequence) over one water and one methanol molecule, <i>i.e.</i> :                                                                                                              | <b>1</b> |
|      | <b>M3</b> either $\delta^- \text{O} - \text{H}^{\delta+} - \cdots \delta^- \text{O}$ or $\text{H}^{\delta+} - \cdots \delta^- \text{O} - \text{H}^{\delta+}$<br>lone pair of electrons on O of H-bond, in line with H-bond | <b>1</b> |

Q55

9701/22/O/N/23/Q4(a)(iii)

Intermolecular Forces

|           |                                                                                                                                                                                                          |          |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4(a)(iii) | dashed (or distinctly labelled) bond between O and H in different molecules of H-O-H and propanoic acid                                                                                                  | <b>1</b> |
|           | correct sequence of three partial charges over the H-bond<br>$\text{H}^{\delta+} \quad \text{O}^{\delta-} - \text{H}^{\delta+}$<br>OR $\text{O}^{\delta-} \quad \text{H}^{\delta+} - \text{O}^{\delta-}$ | <b>1</b> |
|           | lone pair on O in line with H-bond                                                                                                                                                                       | <b>1</b> |

Q56

9701/22/M/J/23/Q1(b)(iv)–(v)

Intermolecular Forces

|          |                                                                                                                                                                                                                                                                                                                                                                                                           |          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1(b)(iv) | <b>M1</b> link shown as a dashed line between the lone pair of electrons from N of one $\text{NH}_3$ to one H on other $\text{NH}_3$<br><b>M2</b> minimum 3 correct partial charges (on adjacent atoms) over two $\text{NH}_3$ molecules<br><b>EITHER</b> $\delta^- \text{N} - \delta^+ \text{H} - \cdots \delta^- \text{N}$ <b>OR</b> $\delta^+ \text{H} - \cdots \delta^- \text{N} - \delta^+ \text{H}$ | <b>2</b> |
| 1(b)(v)  | <b>M1</b> O is more electronegative than N<br><b>M2</b> two H-bonds per water molecule : 1 per ammonia molecule.                                                                                                                                                                                                                                                                                          | <b>2</b> |

Q57

9701/22/O/N/22/Q2(e)(ii)

Intermolecular Forces

|          |                                                                                                                                                                                                                                                                                                                      |          |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(e)(ii) | <b>M1</b> sum of instantaneous dipole – dipole attractions and permanent dipole attractions in $\text{SCl}_2$ is greater than those in $\text{SBr}_2$<br>OR<br>permanent dipole-permanent dipole (forces) in $\text{SCl}_2$ are stronger<br>OR<br>intermolecular forces are stronger in $\text{SCl}_2$<br><b>ora</b> | <b>1</b> |
|          | <b>M2</b> $\text{SCl}_2$ (molecule / bond) is more polar than $\text{SBr}_2$<br>OR<br>Cl is more electronegative than Br so $\text{SCl}_2$ has stronger (permanent) dipoles <b>ora</b>                                                                                                                               | <b>1</b> |

Q58

9701/23/M/J/21/Q2(b)(iii)

Intermolecular Forces

|           |                                                                                                                                                                                                                                                                                                                                    |          |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(b)(iii) | <b>M1</b> (liquid) H <sub>2</sub> O molecules are held by stronger <b>hydrogen bonding</b><br><b>OR</b> it takes more energy to break the <b>hydrogen bonds</b> between water molecules (in the liquid state)<br><b>OR</b> each water molecule forms two <b>hydrogen bonds</b> (whereas methanol can (only) form one per molecule) | <b>1</b> |
|           | <b>M2</b> fewer H <sub>2</sub> O liquid molecules (able to) escape / become gaseous <b>ora</b>                                                                                                                                                                                                                                     | <b>1</b> |

Q59

9701/21/O/N/25/Q1(c)(iv)

Intermolecular Forces

|          |                                                                                                                                                                                                                                                     |          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1(c)(iv) | <b>M1</b> CO <sub>2</sub> has only instantaneous / temporary dipole–induced dipole forces<br><b>AND</b> methanal has permanent dipole–permanent dipole (and id-id) forces<br><b>M2</b> IMF's in methanal are stronger than IMF's in CO <sub>2</sub> | <b>2</b> |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|

Q60

9701/21/O/N/24/Q3(d)(i)

Intermolecular Forces

|         |                                                                                                                                                                                                                                   |          |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(d)(i) | <b>M1</b> molecules CH <sub>4</sub> to PbH <sub>4</sub> have greater number of electrons<br>/ number of electrons increase                                                                                                        | <b>1</b> |
|         | <b>M2</b> greater / stronger instantaneous dipole–induced forces / London forces / dispersion forces<br><b>OR</b><br>more energy required to overcome the instantaneous dipole–induced forces / London forces / dispersion forces | <b>1</b> |

